

llmXive Automated Review of The Mirage of Optimizing Training Policies: Monotonic Inference Policies as the Real Objective for LLM Reinforcement Learning

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper makes several strong claims regarding the theoretical guarantees and empirical metrics that are not fully supported by the provided text or tables.

First, the Abstract and Introduction repeatedly assert that the proposed method ensures “monotonic” improvement of the inference policy. However, Section 3.2 (“Step 2: Inference-Gap-Aware Update Acceptance”) explicitly contradicts this by stating: “This mechanism does not provide a formal monotonic-improvement guarantee, but it reduces the risk of accumulating updates...” This is a significant definitional drift where the term “monotonic” is used in the headline claims to imply a guarantee that the authors themselves admit does not exist. The claims in the Abstract and Introduction must be softened to reflect that the method *aims* for monotonicity or *reduces the risk* of non-monotonicity, rather than ensuring it.

Second, the derivation of the post-update gap proxy $\hat{T}_{\text{post}}$ in Section 3.2 relies on a “reverse identity” to approximate an expectation under the training policy using samples from the inference policy. The text claims this yields a “stable proxy” but does not cite the specific identity used nor provide any theoretical bound on the variance or bias introduced by the importance weighting ρ_i . Without this support, the claim of stability is an overstatement of the evidence provided.

Third, there is a direct contradiction regarding the evaluation metric in Table 1. The table caption explicitly states “Pass@1 accuracy (%)”, yet Section 4.1 (“Evaluation”) states: “For the small benchmarks AIME24 and AMC23, we use avg@16 to reduce evaluation variance.” The table does not indicate which metric is reported for these specific columns. If the table reports Pass@1, the text is misleading; if it reports avg@16, the caption is incorrect. This ambiguity prevents the reader from accurately interpreting the reported gains.

Finally, the specific dataset sizes used for training (5,759 and 1,491 examples) are presented as filtered subsets of larger public datasets (DAPO-Math-17k and DeepMath-103k). The paper claims these subsets were chosen to ensure “meaningful reward variation,” but the specific filtering criteria (e.g., success rate thresholds, difficulty filters) are not described. This omission makes the claim that the training set is optimally constructed for the experiment unsupported.

Action items.

- **[writing]** The paper makes several strong claims regarding the theoretical guarantees and empirical metrics that are not fully supported by the provided text or tables. First, the Abstract and Introduction repeatedly assert that the proposed method ensures “monotonic” improvement of the inference policy. However, Section 3.2 (“Step 2: Inference-Gap-Aware Update Acceptance”) explicitly contradicts this by stating: “This mechanism does not provide a formal monotonic-improvement guarantee, but it reduces the

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure presents four panels comparing methods, but the caption omits the ‘Baseline’ method shown in the first two plots. Additionally, the Clip Ratio chart lacks a y-axis label, though the data is legible.

Figure 2

The figure effectively displays the sensitivity curves, but the rightmost panel lacks a y-axis label for the ‘ T_{post} ’ metric mentioned in the caption, and the ‘Ours’ legend entry is undefined.

Figure 3

The figure presents four ablation panels with a shared legend, but the color scheme for the two orange variants is ambiguous. Additionally, the ‘Inference Gap’ plot visually contradicts the caption’s claim of a ‘controlled’ trajectory for the full method.

Figure 4

The figure presents two line plots with clear axes and a legend, but the caption is insufficient as it fails to map the legend entries (model names) to the specific curves shown, leaving the reader to guess which line represents which model.

Figure 5

Figure 5 effectively visualizes the MIPI principle and the MIPU workflow, clearly distinguishing between canonical RL and the proposed method. The diagrammatic flow, mathematical decomposition, and color coding (blue/red) are well-aligned with the detailed caption, making the conceptual argument easy to follow.

Action items.

- **[science]** Figure 1: The caption claims to compare ‘PPO-IS, Vanilla-IS, and TIS’, but the ‘Reward’ and ‘Mismatch-K3’ plots include a ‘Baseline’ series (grey) that is not mentioned in the caption text.
- **[science]** Figure 1: The ‘Clip Ratio’ bar chart displays a value of 7.18e-03 for TIS, which is an order of magnitude higher than the other methods ($\sim 2e-04$), yet the y-axis is not labeled

with units or a title, making the scale interpretation ambiguous.

- **[science]** Figure 2: The caption claims to show sensitivity to c and T_{post} , but the legend only defines $c = 0$, $c = 0.0001$, $c = -0.0001$, and ‘Ours’. The ‘Ours’ method is undefined in the caption or legend, and the T_{post} metric is not explicitly labeled on any axis (though the rightmost panel is likely it).
- **[writing]** Figure 2: The rightmost panel’s y-axis label is missing; the caption identifies this metric as ‘ T_{post} ’, but the plot itself lacks this label, relying solely on the caption for identification.
- **[science]** Figure 3: The ‘Inference Gap’ panel shows the ‘ours’ method (yellow) oscillating around 0.0000, while the caption claims the full method obtains a ‘more controlled inference-policy trajectory.’ The visual data suggests the ‘ours’ method is unstable or noisy rather than controlled compared to the baselines.
- **[writing]** Figure 3: The legend uses two distinct shades of orange for ‘+ step 1’ and ‘+ step 2’ which are visually very similar and difficult to distinguish from one another in the plots, especially in the ‘Mismatch-K3’ panel.
- **[science]** Figure 4: The caption describes the plot as ‘Inference-training K3-KL and inference gap’ but does not define the colors. The legend identifies the lines as ‘Qwen3-4B FP8’ and ‘Qwen3-1.7B FP8’, yet the caption fails to specify which model corresponds to the orange or red lines, making the data interpretation ambiguous.
- **[writing]** Figure 4: The x-axis label ‘Training Step’ is present, but the caption does not specify the total number of steps or the context of the training run (e.g., which specific experiment or baseline from Figure 3 this data is drawn from).

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper generally maintains a high standard of technical clarity for an audience familiar with Reinforcement Learning and LLMs. Most core concepts like “training-inference mismatch,” “GRPO,” and “importance sampling” are either standard or well-contextualized. However, there are several instances where specific notation, acronyms, or named variants are introduced without immediate, self-contained definitions, forcing the reader to rely on external citations or later sections to understand the precise operational meaning.

Specifically, the notation for visitation distributions (d_{τ}^{π}) in the Preliminaries is introduced without a defining clause, which, while standard in some subfields, is not universal enough to be assumed without definition in a paper proposing a new objective. Similarly, the acronyms “PPO-IS” and “Vanilla-IS” are used as if they are standard terms, but they appear to be specific

labels for the variants discussed in this work (or the cited mismatch1 paper); they require a brief operational definition upon first use to ensure the reader understands exactly what mechanism they refer to (e.g., clipping the full ratio vs. the update ratio).

Furthermore, the truncation notation $\bar{w}$ and the sequence ratio ρ_i are used in equations before the specific truncation bound or the definition of the sequence length T_i is explicitly tied to the symbol in the immediate text. Finally, the term “K3-KL” is used as a metric name without explaining the “K3” component, which is likely a specific approximation order or estimator name not immediately obvious to a general RL researcher. Addressing these points by adding brief parenthetical definitions or one-sentence glosses at the point of first use will ensure the paper is fully self-contained for a competent adjacent-field reader.

Action items.

- **[writing]** The paper generally maintains a high standard of technical clarity for an audience familiar with Reinforcement Learning and LLMs. Most core concepts like “training-inference mismatch,” “GRPO,” and “importance sampling” are either standard or well-contextualized. However, there are several instances where specific notation, acronyms, or named variants are introduced without immediate, self-contained definitions, forcing the reader to rely on external citations or later sections to understand the

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a logically consistent argument: it identifies a mismatch between training and inference policies, proposes a decomposition of the inference improvement objective (MIPI), and validates a two-step method (MIPU) designed to satisfy that decomposition. The ablation studies and analysis sections generally support the claim that the two steps address distinct failure modes.

However, two minor logical ambiguities in the text require clarification to ensure the argument is watertight:

1. **Section 4.3 (Analysis of Step 2):** The text states random rollback is “more conservative” because it rejects 67% of updates compared to Step 2’s 53.5%. While the numbers support the claim, the phrasing could be misread. Explicitly stating “rejects a higher percentage” would remove any ambiguity about the definition of “conservative” in this context.
2. **Appendix (Implementation Details):** The parameter $c_{end} = 0$ for Qwen3-4B implies a strict non-negative acceptance rule. The main text should explicitly define $c = 0$ as “strict non-regression” to ensure readers correctly interpret the tolerance behavior at the end of the annealing schedule.

These are minor phrasing issues that do not invalidate the core logic but should be fixed for

clarity.

Action items.

- **[writing]** In Section 4.3, the text claims random rollback is ‘more conservative’ because it rejects 67% vs Step 2’s 53.5%. While numerically true, the phrasing is slightly ambiguous. Clarify to ‘rejects a higher percentage’ to ensure the logical link between ‘conservative’ and the rejection rate is explicit.
- **[writing]** In the Appendix, `c_end=0` for Qwen3-4B implies strict non-negative acceptance. Ensure the main text explicitly defines `c=0` as ‘strict non-regression’ to prevent misinterpretation of the tolerance parameter’s behavior at the end of annealing.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes a compelling case for objective misalignment in LLM RL, but the rhetoric occasionally outpaces the specific experimental conditions. The abstract and introduction present the “Monotonic Inference Policy Improvement” (MIPI) principle and the MIPU framework as a general solution to training-inference mismatch. However, the empirical validation (Section 5) is exclusively conducted under “FP8-quantized rollout,” a specific high-mismatch scenario. While the authors acknowledge in the Limitations that they are constrained to “moderate-scale models,” the main narrative frames the results as a general improvement in “training stability” and “reasoning performance” without sufficiently qualifying that these benefits were observed specifically in the context of quantization-induced mismatch.

For instance, the abstract states MIPU “improves average reasoning performance and training stability” without the qualifier “under high mismatch.” Similarly, the conclusion reiterates that MIPU improves stability, which, while true for the tested setting, risks implying robustness across all RL training regimes (e.g., full-precision or different mismatch sources) where the method was not tested. The paper would benefit from tightening the scope of its claims to match the specific “FP8-quantized” evidence, or explicitly stating that the method is a targeted solution for low-precision inference environments rather than a universal fix for all training-inference mismatches. The current phrasing suggests a broader generalization than the single-condition experiment supports.

Action items.

- **[writing]** Abstract claims MIPU ‘improves average reasoning performance’ generally, but experiments (Sec 5.1) are restricted to FP8-quantized rollout on two Qwen3 models. Scope the claim to ‘under high training-inference mismatch (e.g., FP8 quantization)’ or add evidence from full-precision settings.
- **[writing]** Conclusion states MIPU improves ‘training stability’ generally, but Limitations

admit tests are only on ‘moderate-scale models’ and never in non-quantized settings. Clarify that stability is demonstrated specifically under quantization-induced mismatch, not as a universal property.

- **[writing]** Introduction frames the solution as addressing ‘training-inference mismatch’ generally, yet validation is only in ‘FP8-quantized rollout’ (Sec 5.1). Clarify that the method targets mismatch ‘amplified by low-precision inference’ rather than all forms of mismatch.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a methodological contribution to Reinforcement Learning for Large Language Models (LLMs), specifically addressing training-inference mismatch in mathematical reasoning tasks. The work involves training models (Qwen3-1.7B and Qwen3-4B) on public mathematical benchmarks (MATH-500, AIME24, etc.) using standard RL algorithms (GRPO) and a proposed variant (MIPU).

From a safety and ethics perspective, the research is low-risk. The datasets used are standard, public, and non-sensitive mathematical problem sets. The models are open-weight (Qwen3) and trained on these benign tasks. The proposed method (MIPU) aims to improve training stability and performance; it does not introduce capabilities for generating harmful content, bypassing safety filters, conducting cyberattacks, or deceiving users. The “risk” discussed in the paper is purely technical (training collapse or performance degradation), not societal harm.

The paper includes a “Licenses” section (Appendix) that correctly identifies the licenses for the datasets and models used (MIT, Apache-2.0, etc.), demonstrating compliance with data provenance norms. There is no use of human-subject data requiring IRB approval, no release of Personally Identifiable Information (PII), and no dual-use capabilities described that would lower the barrier to a specific harmful application (e.g., automated vulnerability discovery or biological synthesis).

Consequently, there are no foreseeable, non-trivial risks of harm that the paper fails to acknowledge or mitigate. The standard “broader impacts” discussion is not strictly required for this type of algorithmic optimization paper, and the absence of a dedicated ethics section does not constitute a safety failure given the benign nature of the data and task. The paper is safe to accept.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling theoretical argument for objective misalignment in LLM RL and proposes MIPU to address it. However, the empirical evidence supporting the central claims of performance improvement and stability is currently insufficient to rule out sample noise or lucky seeds.

The primary concern lies in the lack of variance reporting. Table 1 presents headline accuracy numbers (e.g., 66.71% vs 65.66% for Qwen3-4B) derived from what appears to be single runs. In reinforcement learning, especially with small training sets (1,491 examples for Qwen3-4B) and high-mismatch settings, performance variance across seeds is often substantial. A 1.05% average gain is well within the plausible range of sampling noise for a single run. Without reporting results across multiple seeds (e.g., 3-5) with standard deviations or confidence intervals, the reader cannot distinguish a genuine algorithmic improvement from a lucky initialization or a specific random seed trajectory. The claim of “improved average reasoning performance” is statistically unsupported in its current form.

Similarly, the claim of “training stability” relies heavily on the visual inspection of a single training curve (Figure 1) and a binary “Stable” flag in Table 1. While the curve shows MIPU avoiding a sharp drop seen in baselines, this observation is anecdotal without statistical backing. Does the baseline *always* collapse, or did it just collapse in this specific run? To substantiate the stability claim, the authors should either report the variance of the final performance across seeds (showing MIPU has lower variance) or define a quantitative metric for stability (e.g., maximum performance drop during training) and show it is significantly better for MIPU across multiple runs.

Finally, the ablation study in Table 2 has a subtle confound in the “Step 2 only” condition. The text implies this variant uses the standard GRPO update as its proposal, meaning the “Step 2 only” result is a combination of a potentially poor proposal (GRPO) and the Step 2 filter. This makes it difficult to isolate the specific contribution of the Step 2 acceptance mechanism. A more rigorous ablation would compare the full MIPU against a variant that uses the *same* Step 1 proposal (TIS) but replaces the inference-gap-aware acceptance with a random or fixed-threshold acceptance rule. This would better isolate whether the gain comes from the *specific* inference-gap signal or simply from the act of rejecting updates more frequently.

Addressing these points by adding multi-seed runs and refining the ablation design is necessary to transform the current suggestive results into robust scientific evidence.

Action items.

- **[writing]** The paper presents a compelling theoretical argument for objective misalignment in LLM RL and proposes MIPU to address it. However, the empirical evidence supporting the central claims of performance improvement and stability is currently insufficient to rule out sample noise or lucky seeds. The primary concern lies in the lack of variance reporting. Table 1 presents headline accuracy numbers (e.g., 66.71% vs 65.66% for Qwen3-4B) derived from what appears to be single runs. In reinforcement lear

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical reporting in this paper lacks the necessary uncertainty quantification required to validate the comparative claims made. While the experimental design (FP8 rollout, specific benchmarks) is described, the results in **Table 1** (Main Results) and **Table 2** (Ablation) present performance metrics (pass@1 accuracy) as precise point estimates (e.g., 66.71%, 53.97%) without any accompanying standard deviation, standard error, or confidence intervals. In reinforcement learning, where training dynamics are stochastic and sensitive to random seeds, reporting a single number (or a number derived from an unspecified number of runs) creates an illusion of precision that the data does not support.

Specifically, the claim that MIPU is “more stable” is supported by a binary “Stable” column in Table 1 and visual inspection of **Figure 1**, rather than a quantitative metric of variance or a statistical test of the training trajectory’s volatility. Without reporting the variance across multiple independent runs (seeds), it is impossible to determine if the observed differences between MIPU and baselines (e.g., the 1.05% gain on Qwen3-4B) are statistically significant or merely noise. The text uses terms like “significantly better” and “stable” without the statistical machinery (p-values, effect sizes, or variance measures) to back them up.

To rectify this, the authors should report results as mean $\pm$ standard deviation over at least 3-5 independent seeds for all methods. If the computational cost prohibits multiple seeds, the authors must explicitly state that results are from a single run, remove claims of statistical significance, and replace the binary “Stable” column with a quantitative metric of training stability (e.g., the standard deviation of the score over the final 20% of training steps).

Action items.

- **[writing]** Table 1 and Table 2 report pass@1 accuracy to two decimal places (e.g., 66.71%) without any measure of uncertainty (SD, SE, or CI) or mention of the number of random seeds used. In RL experiments, performance is highly stochastic; reporting a single point estimate implies false precision. Report mean $\pm$ SD over at least 3 independent seeds for all methods, or explicitly state that results are from a single run and rephrase claims of ‘stability’ accordingly.
- **[writing]** The ‘Stable’ column in Table 1 is a binary qualitative judgment (checkmark/cross) derived from visual inspection of training curves (Figure 1) rather than a statistical test of variance or a formal stability metric (e.g., coefficient of variation over the final 20% of training). Define a quantitative stability metric (e.g., max drop from peak, or variance of the last 100 steps) and report it numerically to support the binary classification.
- **[writing]** The paper claims MIPU is ‘significantly better’ or ‘more stable’ than baselines in the abstract and conclusion, but no hypothesis tests (e.g., paired t-tests or Wilcoxon signed-rank tests) are reported in the text or tables to support these comparative claims.

If multiple seeds are run, perform a statistical test on the final scores and report p-values; if only one run is available, remove the word ‘significantly’ and qualify the comparison as ‘observed improvement’.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a clear and compelling argument regarding the training-inference mismatch in LLM reinforcement learning. The logical flow from the problem statement to the proposed MIPI objective and the MIPU framework is generally strong. However, there are several specific instances where the prose requires polishing to ensure the reader can parse sentences on the first pass without stumbling over grammatical errors or ambiguous phrasing.

The most significant issue is a grammatical error in Section 3.1 (Methodology), where a sentence describing PPO-IS is constructed as a run-on: “we refer to this direct sampler-referenced variant as PPO-IS, instead clips the full trainer-to-sampler ratio.” This structure forces the reader to re-parse the clause to understand that “clips” is the verb for the variant, not a new independent clause. A simple insertion of “which” fixes this.

Additionally, there are minor typos and citation formatting inconsistencies that disrupt the reading experience. In Appendix A.1, the word “truncted” appears instead of “truncated.” In the same appendix, the phrase “Following the authors’ guidance” is vague; without a specific citation, the reader is left guessing which prior work established the specific decay intervals mentioned. In Section 3.2, the use of the LaTeX symbol `\leadsto` within the sentence flow is slightly jarring; while mathematically precise, standard prose words like “approximates” or “yields” might flow better for a general reader, or the citation style for `mismatch1` needs to be consistent (using `\citet` rather than raw text) throughout the document.

Finally, in Appendix A.2, the sentence referencing “mismatch1” lacks the proper LaTeX citation command, appearing as raw text. While these are minor mechanical issues, they accumulate to create friction for the reader. Correcting these specific sentences and ensuring consistent citation formatting will significantly improve the overall readability and professionalism of the paper. The core scientific narrative is sound and well-structured, but these surface-level errors should be addressed before publication.

Action items.

- **[writing]** The manuscript presents a clear and compelling argument regarding the training-inference mismatch in LLM reinforcement learning. The logical flow from the problem statement to the proposed MIPI objective and the MIPU framework is generally strong. However, there are several specific instances where the prose requires polishing to ensure the reader can parse sentences on the first pass without stumbling over grammatical errors or ambiguous phrasing. The most significant issue is a grammatical err